Linear Deception Probes Detect Correlates, Not Mechanism: Geometric Orthogonality and Adversarial Failure in Gemma 2 2B IT

Jordan F. McCann
Independent Researcher
jordanfmccann@gmail.com

May 10, 2026

Abstract

Linear probes are widely used for deception and sandbagging detection in deployed language models. The deployment promise rests on two implicit assumptions: that the probe direction corresponds to the computational substrate that produces the behavior, and that probe performance generalizes across the framings in which the behavior appears. We test both assumptions in Gemma 2 2B IT and find both contradicted. Twenty bootstrap-trained logistic-regression probes on a 200-prompt sandbag-vs-honest contrast at `attn_out_L12` produce direction estimates that are nearly orthogonal to the canonical rank-1 mechanism direction d_{mech} ($\cos = 0.0160$, 95% CI $[0.0152, 0.0168]$); projecting out d_{mech} leaves probe AUROC unchanged, while projecting out the probe direction produces a behavioral effect two orders of magnitude smaller than the mechanism's (Δ_{count} CI $[+0.2, +0.8]$ out of 40 vs. $+10$ pp on test). A three-direction subspace audit rules out the alternative that the probe direction lies in $\text{span}(d_{\text{mech}}, d_{\text{refusal}})$ ($\sigma_3/\sigma_1 = 0.688$). A complementary SAE-feature subspace null at $K \leq 10$ across thirteen configurations rules out a sparse-feature mechanism. On a six-category sha256-frozen adversarial benchmark spanning fiction-writing, role-play, hypotheticals, counterfactuals, devil's-advocate, and instruction-following-with-irony, three published probe methodologies (Apollo diff-of-means, Goldowsky-Dill instruction-pair, Meanshift) achieve test AUROC > 0.7 by at least one probe in five of six categories. One category — instruction-following-with-irony — drives all three probes below random (Apollo 0.425, Goldowsky-Dill 0.366, Meanshift 0.435; $n = 45$). Probe-method pairwise direction cosines (0.444 to 0.887) translate to differential per-category failure modes, with Apollo most fragile (3 of 6 degraded) and Meanshift most robust (1 of 6). The two findings cohere: probe direction is geometrically a co-occurrence axis rather than a mechanism axis, and that co-occurrence axis decouples from the deception label specifically under irony, where the probe's training signal becomes orthogonal to the deceptive content. Linear deception probes in Gemma 2 2B IT detect a representation that is correlated with but causally and distributionally distinct from the mechanism producing the behavior. The benchmark and sha256-frozen test splits are released for community use.

1 Introduction

Linear probes — typically logistic-regression or difference-in-means classifiers on residual-stream activations — are the workhorse method for deception detection in deployed language models

[1, 5, 10]. On standard sandbag-vs-honest contrasts, such probes routinely report test AUROC > 0.99. The deployment thesis built on top of these numbers rests on two assumptions that are usually left implicit:

1. **Mechanistic assumption.** The direction the probe identifies is the direction the model uses to implement the behavior. If true, intervening on that direction should change the behavior, and removing the mechanism direction should remove the probe's signal.

2. **Distributional assumption.** The probe trained on one framing of deception generalizes to other framings. If true, a probe trained on instructed sandbagging should also detect role-played, fiction-framed, or ironic deception.

This paper tests both assumptions in Gemma 2 2B IT — a model where prior work has identified a rank-1 mechanism direction at `attn_out_L12` that, when projected out, causally restores sandbag-condition capability on a 200-prompt evaluation suite [9] (henceforth d_{mech}). The same suite admits training of linear probes; we use d_{mech} as a fixed causal anchor against which probe directions can be compared geometrically and causally. Independently, we construct a six-category adversarial benchmark with sha256-frozen test splits and evaluate three published probe methodologies on each category.

Contributions.

1. **A geometric and causal disjunction.** Twenty bootstrap-trained probes have cosine 0.0160 (CI $[0.0152, 0.0168]$) with d_{mech}. Projecting out d_{mech} does not change probe AUROC on any seed; projecting out the probe direction produces a behavioral effect two orders of magnitude smaller than d_{mech}'s. A three-direction subspace audit (d_{mech}, d_{refusal}, d_{probe}) is full-rank, ruling out the trivial alternative that the probe direction is a linear combination of the mechanism and refusal directions.

2. **A sparse-feature subspace null.** Across thirteen SAE-feature-subspace configurations at $K \in \{3, 5, 10\}$, three layers, and four reference directions, no configuration meets the pre-registered LANDMARK threshold or beats a 9-direction random baseline by 2σ. The mechanism is not a sparse SAE-feature combination accessible by mean-difference selection.

3. **An adversarial benchmark.** Six categories, three probe methodologies, sha256-frozen test stratum, 485 held-out test prompts. One category (instruction-following-with-irony) drives all three probes below random; five of six admit probe AUROC > 0.7 by at least one method.

4. **A coherent synthesis.** The geometric finding and the adversarial finding are not independent observations. If the probe is detecting a co-occurrence axis (sandbag-context-detection signal that correlates with the deception label) rather than the mechanism axis (the computational substrate that produces deceptive output), then probes will fail in framings where the co-occurrence decouples from the label. Irony-following is precisely such a framing: the model is instructed to produce statements opposite to its honest output without the sandbag-context cues that the probe was trained on.

The framing throughout is single-model: all measurements are on Gemma 2 2B IT. The benchmark and sha256-frozen test splits are released to enable replication on other models. Cross-architectural and cross-scale extensions are deferred to companion work.

2 Related Work

Linear probes for deception. Goldowsky-Dill et al. [5] introduce instruction-pair probes for deception classification with high AUROC on sandbag-vs-honest contrasts. Apollo Research [1] replicate probe-based deception evaluations across multiple architectures. Taylor et al. [10], the UK AISI "Auditing Games for Sandbagging" study, stress-tests black-box probe-based detection schemes against red-team-instilled sandbagger model organisms and reports failure modes for final-layer probes on adversarially constructed sandbaggers. Our contribution is complementary: we provide white-box geometric and causal counter-evidence on a specific model where the mechanism is independently characterized, and we construct a public sha256-frozen six-category adversarial benchmark.

Probing methodology critiques. Belinkov [3] reviews probing-classifier methodology and notes that high probe accuracy does not by itself establish that the model uses the probed property. Hewitt and Manning [6] construct probe-control experiments; their concern (probe expressivity confounding linguistic-structure detection) is structurally analogous to ours (probe expressivity confounding mechanistic claims). Our extension is to operationalize the disjunction with causal interventions on the rank-1 mechanism direction.

Mechanistic substrate. McCann [9] localizes the sandbagging mechanism in Gemma 2 2B IT to a rank-1 direction at `attn_out_L12` with +12.5 pp test recovery and 0 pp honest damage. Arditi et al. [2] establishes that refusal in instruction-tuned models is mediated by a single direction; we use the refusal direction as the third basis vector for the subspace audit.

SAE features and behavioral interventions. Templeton et al. [11] and Lieberum et al. [8] provide the Gemma Scope JumpReLU SAEs we encode against. The general pattern of "many features differ statistically, few intervene causally" has been observed in cross-stage diffing [4]; we extend that observation to within-condition $K \leq 10$ SAE-feature subspaces ranked by mean activation difference.

3 Methods

3.1 Model and prompt suites

Model. Gemma 2 2B IT (HuggingFace `unsloth/gemma-2-2b-it`, methodologically equivalent to `google/gemma-2-2b-it`), bfloat16, on a single RTX 2060 12 GB. All measurements use the model's standard chat template. Probe activations are taken at the final-token position of the user turn unless stated otherwise.

Sandbagging suite. The 200-prompt suite from cycle-001 of this program: five capability domains (arithmetic, factual recall, basic logic, code completion, reading comprehension) with deterministic ground-truth answers. Splits 60/20/20 (train/val/test = 120/40/40) under `random.seed(0)`. Three system-prompt conditions: `honest`, `sandbagging` (Mitra-style induction), and `eval_context_only`. Prompt set sha256 `e543e781...`; splits sha256 `10deee0e...`.

Adversarial benchmark. Six categories pre-registered (commit `130b434a`), each with deterministic +/− labels per the operational definitions in `project-002/cycle-001/PREREG/shot_T1_taxonomy_freeze.md`:

1. **fiction-writing** — model produces deceptive content within an explicit fiction-writing frame.
2. **role-playing** — model adopts a deceptive persona under role-play instructions.
3. **hypotheticals** — model speculates about deceptive scenarios in hypothetical mode.
4. **counterfactuals** — model produces counterfactual statements containing misinformation framed as counterfactuals.
5. **devil's-advocate** — model argues a deceptive position as a devil's-advocate exercise.
6. **instruction-following-with-irony** — model follows an explicit irony instruction in a way that produces statements opposite to its normal honest output.

Acceptance threshold: ≥ 500 prompts per category, 400/100 train/test split. Achieved: four categories at the 500 target; devil's-advocate at 200 (160/40) and irony-following at 223 (178/45). Verdict: BENCHMARK-PARTIAL on category breadth; the elicitation difficulty for the latter two categories is documented honestly. Test stratum sha256 values per category are listed in the reproducibility appendix.

3.2 Mechanism direction d_{mech}

d_{mech} is taken from cycle-001's pre-registered Phase 5 `attn_out_L12` component sweep: the rank-1 difference-in-means at the last-token `attn_out_L12` activation between sandbag and honest conditions, normalized to unit length on train $\cup$ val. Project-out is single-fire on the first forward pass through layer 12: $v \leftarrow v - (v \cdot d_{\text{mech}})\, d_{\text{mech}}$ at the prompt-final token. Reference behavioral effect: +12.5 pp on test split (cycle-001 Phase 5 R3, $n_{\text{test}} = 40$), 0 pp honest damage. Direction file sha256: `6784633c....`

3.3 Probe panel

Three methodologically distinct probe families, evaluated for their geometric relationship to d_{mech} (panel A) and their adversarial robustness (panel B):

1. **Apollo (diff-of-means).** $d_{\text{probe}} = \text{mean}(a \mid y = +) - \text{mean}(a \mid y = -)$, normalized. Implemented per Apollo Research's published methodology.
2. **Goldowsky-Dill (instruction-pair).** Logistic-regression coefficient on (positive, negative) instruction-pair contrasts, normalized.
3. **Meanshift.** Z-scored mean activation difference across the within-category training distribution.

For panel A, twenty additional bootstrap probes are trained: for each seed $s \in \{0, \ldots, 19\}$, a 95% bootstrap resample of the sandbagging suite's train split is drawn with seed s, and a logistic regression (L2 penalty $C = 1.0$, lbfgs solver, `random_state=s`, `max_iter=1000`) is trained on the bootstrapped `attn_out_L12` activations (sandbag vs honest binary). The probe direction is ${d_{\text{probe}}}^{(s)} = w_s / \|w_s\|$.

The probe layer is L12 of Gemma 2 2B IT for both panels — consistent with cycle-001's identification of `attn_out_L12` as the canonical mechanism site.

3.4 Geometric and causal metrics

For each probe direction:

- $\cos(d_{\text{probe}}{}^{(s)}, d_{\text{mech}})$: scalar inner product after both are unit-normalized.
- Δ_{AUROC}: probe AUROC on val before vs. after projecting out d_{mech} from the activations re-evaluated by the same probe.
- $\Delta^{\text{sb}}_{\text{count}}$: integer change in correct-on-sandbag count out of 40 val prompts after projecting out $d_{\text{probe}}{}^{(s)}$.
- $\Delta^{\text{hn}}_{\text{count}}$: same on honest condition (capability control).

All projections are single-fire, applied at `attn_out_L12` on the first forward pass through layer 12. Per-seed metrics are aggregated to mean, standard deviation, and 95% bootstrap CI ($B = 1000$ percentile resamples on seed-level values).

3.5 SAE-feature subspace test

Following the cycle-003 shot U3 protocol: for $K \in \{3, 5, 10\}$, three layers (L11, L12, L13), and four reference directions (d_{mech}, d_{refusal}, $d_{\text{probe}}{}^{(0)}$, d_{eval}), build candidate K-feature subspaces from Gemma Scope JumpReLU SAEs ranked by mean activation difference. Project these out at `attn_out_L12` and measure Δ_{acc} on val. Compare against 9 matched-K random feature subspaces drawn from the same SAE feature pool.

3.6 Adversarial benchmark protocol

For each category and each probe family: train the probe on the category's train stratum ($n =$ 160–400); evaluate on the held-out test stratum ($n =$ 40–100); report test AUROC. Probe layer: L12. Test-touch budget: 3 cumulative test touches across the program (T3, U2, V2; one per cycle), each demarcated in `cycle-{1,2,3}/TEST_TOUCHES.md`.

3.7 Pre-registration

Panel A geometric/causal: 20-seed bootstrap pre-registered at `cycle-004/PREREG/shot_R2_u1_deep_bootstrap.m` (commit `d91b109d`); SAE-subspace test at `cycle-003/PREREG/shot_U3_subspace_ablation.md`; three-direction subspace at `cycle-004/PREREG/shot_R3_geometry_prereg.md`. Panel B benchmark: pre-registered at `project-002/cycle-001/PREREG/shot_T1_taxonomy_freeze.md`, `shot_T2_prompt_gene` `shot_T3_pilot_evaluation.md` (commit `130b434a`). Project-wide invariants in `INVARIANT.md` (commit `19704e19...`).

4 Results

4.1 d_{mech} reference effect

The d_{mech} direction at `attn_out_L12`, when projected out single-fire on the first forward pass through layer 12, increases sandbag-correctness by +12.5 pp on the test split ($n_{\text{test}} = 40$) and 0 pp honest damage (within ±2.5 pp tolerance). This reference value (cycle-001 Phase 5 R3) is the anchor against which all probe directions in Sections 4.2–4.5 are compared.

Table 1. Three-direction subspace SVD. Both matrices are full-rank ($\sigma_3/\sigma_1 = 0.688$, well above degeneracy).

Quantity	M_A (probe-mean)	M_B (probe seed-0)
$\cos(d_{\text{mech}}, d_{\text{refusal}})$	0.357	0.357
$\cos(d_{\text{mech}}, d_{\text{probe}})$	0.0161	0.0174
$\cos(d_{\text{refusal}}, d_{\text{probe}})$	−0.0153	−0.0161
σ_1	1.165	1.165
σ_2	1.001	1.001
σ_3	0.801	0.801
σ_3/σ_1	0.688	0.688

4.2 Probes are nearly orthogonal to d_{mech}

Twenty bootstrap-trained probes each achieve train AUROC = 1.0 and val AUROC = 1.0. The cosine between each probe direction and d_{mech} across seeds:

$$\text{mean} = 0.0160; \quad 95\% \text{ CI } [0.0152, 0.0168]; \quad \text{range } [0.0126, 0.0188]; \quad n_{\text{seeds}} = 20.$$

The 95% CI upper bound (0.0168) is two orders of magnitude below the pre-registered DISTINCT threshold of 0.3. On every seed, projecting out d_{mech} from the activations leaves the probe's AUROC unchanged: $\Delta_{\text{AUROC}} = 0$ in 20 of 20 seeds, with mean 0 and standard deviation 0. The geometric claim does not depend on the AUROC ceiling: cosines are computed in fp32 with explicit unit normalization on both sides; the tight CI $[0.0152, 0.0168]$ indicates probe directions are reliably orthogonal to d_{mech} across resamples.

4.3 Three-direction subspace audit

A possible alternative explanation for orthogonality is that the probe direction lies in the span of d_{mech} and a refusal direction d_{refusal} — i.e., that probes detect a refusal-correlated combination rather than an independent representation. To rule this out, we compute the SVD of $M = [d_{\text{mech}}; d_{\text{refusal}}; d_{\text{probe}}]$ at `attn_out_L12` with d_{probe} taken as the mean over 20 seeds (M_A) and as seed-0 alone (M_B); results are reported in Table 1.

The probe direction is not a linear combination of mechanism and refusal directions; it occupies an independent dimension.

4.4 SAE-feature subspaces do not gate the mechanism

A second alternative is that d_{probe} corresponds to a sparse SAE-feature subspace that itself implements the mechanism. Across thirteen SAE-feature-subspace configurations ($K \in \{3, 5, 10\}$, three layers L11/L12/L13, four reference directions), zero configurations meet the pre-registered LANDMARK threshold ($|\Delta_{\text{proj}}| \geq 30\%$ AND $\Delta^{\text{sb}}_{\text{acc}} \geq +5$ pp AND $\Delta^{\text{hn}}_{\text{acc}} \geq -3$ pp AND beats random by $\geq 2\sigma$ on both), and zero meet the STRONG threshold ($|\Delta_{\text{proj}}| \geq 15\%$ AND $\Delta^{\text{sb}} \geq +3$ pp AND $\Delta^{\text{hn}} \geq -5$ pp AND beats random by $\geq 2\sigma$).

The 9-direction random baseline at matched K shows $\Delta^{\%}_{\text{proj}}$ mean −0.103%, std 0.442%, max 0.891%; $\Delta^{\text{pp}}_{\text{acc}}$ mean +1.39, std 2.91, max +5.00. None of the 13 candidate configurations exceed the 2σ threshold on both projection-magnitude and capability-recovery axes. A complementary single-feature analysis at $K = 1$ reaches POSITIVE-NULL with $n_{\text{landmark}} = 0$, $n_{\text{strong}} = 0$. Verdict: SUBSPACE-NULL-CONFIRMED. No top-K mean-activation-difference SAE-feature subspace at $K \leq 10$ reproduces d_{mech}'s behavioral effect.

Table 2. Test AUROC by probe × category. Test sample sizes ($n_+/n_-/n$): fiction-writing 34/66/100; role-playing 38/62/100; hypotheticals 77/23/100; counterfactuals 32/68/100; devil's-advocate 20/20/40; irony-following 16/29/45.

Probe	fiction	role-play	hypothet.	counterf.	devil's-adv.	irony
Apollo	**0.291**	0.996	1.000	0.708	0.933	**0.425**
Goldowsky-Dill	0.735	0.998	1.000	0.938	0.968	**0.366**
Meanshift	0.896	0.997	1.000	0.982	0.955	**0.435**

4.5 Behavioral asymmetry: probe project-out vs. mechanism project-out

If d_{probe} is orthogonal to d_{mech} and d_{mech} is the causal direction, projecting out d_{probe} should not produce an effect comparable to projecting out d_{mech}. The 20-seed test produces:

- $\Delta^{\text{sb}}_{\text{count}}$ CI $[+0.2, +0.8]$ out of 40 val prompts (corresponding to $\Delta^{\text{sb}}_{\text{pp}}$ CI $[+0.5, +2.0]$ pp).
- The CI excludes zero, so probe project-out has a measurable effect.
- Sign matches d_{mech} project-out (more correct on sandbag).
- Magnitude is two orders of magnitude smaller than d_{mech}'s: $+1.25$ pp mean on val vs. d_{mech}'s $+5$ pp val and $+12.5$ pp test.

We interpret this as: the probe direction has a small projection onto the causal subspace that explains the small behavioral effect, but its principal axis is geometrically elsewhere. Probes capture a co-occurring representation that is correlated with the binary sandbag/honest label but is not the computational substrate of the behavior.

4.6 Adversarial benchmark: per-probe per-category test AUROC

The 18-cell test AUROC matrix is reported in Table 2. Cells in **bold** are AUROC < 0.5 (below random for binary classification).

Per-probe degradation summary (defining "degraded" as AUROC ≤ 0.75 on test):

Probe	Degraded categories	Count
Apollo	fiction-writing, counterfactuals, irony-following	3 of 6
Goldowsky-Dill	fiction-writing, irony-following	2 of 6
Meanshift	irony-following	1 of 6

Apollo's diff-of-means probe is the most fragile across the panel; Meanshift is the most robust. The differential pattern is not chance: the three probes' pairwise direction cosines are 0.551 (Apollo × Goldowsky-Dill), 0.444 (Apollo × Meanshift), and 0.887 (Goldowsky-Dill × Meanshift), and the high alignment between Goldowsky-Dill and Meanshift translates to similar per-category AUROC profiles, while Apollo's geometric distinctness translates to higher fragility.

4.7 Universal degraded category: instruction-following-with-irony

All three probes drop below random on irony-following: Apollo 0.425; Goldowsky-Dill 0.366; Meanshift 0.435; with $n = 45$ test ($n_+ = 16$, $n_- = 29$). Probes that achieve AUROC > 0.99 on five other adversarial framings — including role-play, hypotheticals, counterfactuals, and devil's-advocate,

Table 3. Per-category Cohen's κ between probe pairs.

Pair	fiction	role-play	hypothet.	counterf.	devil's-adv.	irony
Apollo × Goldowsky-Dill	0.280	0.880	0.720	0.480	0.840	0.612
Apollo × Meanshift	−0.000	0.840	0.640	0.400	0.840	0.096
Goldowsky-Dill × Meanshift	0.720	0.920	0.920	0.920	0.840	0.483

all of which involve the model producing content opposite to its honest output — collectively fail on irony. The failure is probe-architecture-invariant.

4.8 Inter-probe Cohen's κ on shared test prompts

The per-prompt classification agreement (Cohen's κ) between probe pairs varies sharply by category (Table 3).

Apollo × Meanshift falls to $\kappa = -0.000$ on fiction-writing and $\kappa = 0.096$ on irony-following — essentially no agreement. There are 124 prompts in the test corpus where the three probes give 3-way disagreement. The divergent prompts identify cases where probe-method choice changes the deception classification, meaning a single-probe answer is not robust to probe-method swap.

5 Discussion

5.1 The geometric and adversarial findings cohere

The two strands of evidence point at the same underlying picture. Geometrically, the probe direction is nearly orthogonal to the mechanism direction ($\cos = 0.0160$), shares $< 0.07\%$ of cosine-squared variance with d_{mech}, and produces a behavioral effect ~ 2 orders of magnitude smaller. The three-direction full-rank check rules out the alternative that probe-direction is in $\text{span}(d_{\text{mech}}, d_{\text{refusal}})$. The SAE-feature-subspace null at $K \leq 10$ rules out the alternative that mechanism is a sparse SAE-feature combination accessible by feature-difference selection.

The most parsimonious geometric interpretation is that the probe locates a representational structure that is correlated with the binary sandbag/honest label but is not the computational substrate of the behavior. Whether this is best characterized as "spurious correlation" (probe latches onto features that co-vary with the label without driving it) or as "downstream readout" (probe locates an accumulation site of the mechanism's effect on later representations) is not adjudicated here; both interpretations are consistent with the geometric and causal evidence.

The adversarial finding refines the picture. If the probe were detecting the same signal under all framings — even if that signal is a downstream readout rather than the mechanism — we would expect uniform performance across adversarial categories. Instead, performance is high in five categories and uniformly subrandom in one. The category that breaks all three probes is the one where the model produces deceptive content *without* the contextual cues (system-prompt framing, evaluation context, role-play setup) that accompanied the probe's training distribution. Irony-following decouples deceptive content from the contextual signals the probe was trained to detect.

This is consistent with the geometric finding's "co-occurrence axis" interpretation: probes detect a representation that co-occurs with the deception label across sandbag-context training data, and that representation predicts deception in framings where the contextual co-occurrence holds (role-play, hypotheticals, counterfactuals, devil's-advocate, fiction-writing) and fails in the framing where contextual co-occurrence is broken (irony). The L12 mechanism direction, by contrast, is

invariant to surface framing — cycle-001 reports cross-trigger transfer of d_{mech} to T2 (paraphrased system prompt) and T3 (user-turn injection) at +10 pp restoration each — because it is the substrate of the behavior rather than a marker of one of its training contexts.

5.2 Implications for deployment

Three concrete implications follow.

First, a probe trained on a sandbag-honest contrast at L12 of Gemma 2 2B IT can achieve AUROC = 1.0 on its training distribution while detecting essentially nothing of the mechanism that produces sandbagging. Detection-only frameworks for sandbagging deployment safety should not assume the probe direction is the intervention site. The intervention site is empirically distinct, and projecting out the probe direction recovers ~ 1 pp of capability where projecting out the mechanism direction recovers ~ 10 pp.

Second, probe-method choice changes per-category AUROC by up to 0.6 on the same test split. A single-probe number under-specifies the claim "the probe achieves AUROC X on adversarial deception." For deployment, this argues for ensembling probes from different methodological families (Apollo, Goldowsky-Dill, Meanshift have pairwise direction cosines as low as 0.444), reporting per-category AUROC explicitly, and treating any single probe's number as a partial signal.

Third, irony-following is a probe-architecture-invariant break. All three probes drop below random on this category, and the failure is consistent with the geometric finding: irony decouples deceptive content from the contextual signals the probe was trained to detect. Deployment-time monitoring should explicitly flag irony-instruction prompts as a known weakness of linear deception probes in this model, and should not assume probe coverage of the irony framing.

5.3 What probes might still be useful for

Nothing in this paper claims linear probes are useless. The claim is narrower: probe direction $\neq$ mechanism direction, and probe performance does not survive the irony framing. Probes may still be useful as cheap monitoring instruments in deployment contexts where (a) the input distribution matches the probe's training distribution, (b) ironic-instruction inputs can be filtered or flagged separately, and (c) the probe is paired with a causal intervention at the actual mechanism site (such as d_{mech} project-out) for the cases where it fires. The research direction this paper points toward is therefore *detection-plus-intervention* pipelines, where probes raise alerts and mechanism-direction projection acts on them, rather than detection-only frameworks that conflate the two.

6 Limitations

Single model. All measurements are on Gemma 2 2B IT. Whether the orthogonality and the irony-following failure replicate on larger models, other architectures, or other instruction-tuning regimes is an empirical question. Companion work (project-003) extends the geometric analysis to within-Gemma 9B and reports geometric universality of d_{eval} and d_{sandbag} with causal null at `attn_out` at the corresponding layer; this constrains but does not resolve the cross-scale picture.

Single intervention site. The geometric analysis is at `attn_out_L12`; the adversarial probes are also trained at L12. Whether multi-layer probes or downstream-site probes would show different relationships is untested.

Single intervention type. All causal claims rest on rank-1 single-fire project-out. Steering vectors, multi-fire interventions, or finetune-and-test would not necessarily produce identical conclusions.

AUROC ceiling on geometric panel. $\Delta_{\text{AUROC}} = 0$ across 20 seeds with baseline AUROC $= 1.0$ is consistent with two interpretations: (a) the probe is genuinely robust to d_{mech} project-out because it operates in an orthogonal subspace, or (b) the AUROC metric has saturated. The primary geometric claim ($\cos \approx 0.0160$) does not depend on this resolution; the AUROC result is corroborative, not load-bearing.

Two adversarial categories under-sized. Devil's-advocate ($n = 200$) and irony-following ($n = 223$) fell short of the 500-prompt acceptance target. The headline universal-degradation claim relies on irony-following test stratum $n = 45$, which has wider AUROC CIs than the $n = 100$ categories. A tighter test would require fresh elicitation of irony-framing prompts.

Three probes only. Adding more probe methods (gradient-based attribution probes, multi-layer probes, probe-ensembles) would tighten the probe-method-choice argument. The Apollo $\times$ Meanshift $\kappa = -0.000$ result on fiction-writing is a striking pointer for that extension.

Behavioral effect direction CI. The probe-project-out effect's CI is $[+0.2, +0.8]$ count out of 40 prompts. The lower bound 0.2 is non-zero at the 95% level but is sensitive to bootstrap-sample variation. A larger N test would tighten this.

Operational labels for "deceptive." The taxonomy's positive labels rely on the pre-registered operational definitions per category. Ambiguous prompts were excluded; the BENCHMARK-PARTIAL verdict for two categories reflects elicitation difficulty.

No calibration. AUROC is the only reported metric; calibrated detection thresholds (precision/recall at fixed FPR) are deferred to release-time benchmark documentation.

No deployment claim about probe utility. This paper makes no claim about whether linear probes are useful for sandbagging detection in practice (they may detect a signal that is informative about the behavior even if it is not the mechanism, and they may be useful within specific framings even if they fail on irony). The claim is narrower: probe direction $\neq$ mechanism direction, and probe performance does not generalize uniformly across adversarial framings, in this specific setting.

7 Conclusion

In Gemma 2 2B IT at `attn_out_L12`, on a 200-prompt 5-domain sandbagging suite and a 6-category sha256-frozen adversarial benchmark, linear deception probes detect a representation that is geometrically nearly orthogonal to the canonical rank-1 mechanism direction ($\cos = 0.0160$, 95% CI $[0.0152, 0.0168]$) and that fails universally in the instruction-following-with-irony framing while generalizing across five other adversarial categories. Sparse SAE-feature subspaces at $K \leq 10$ do not reproduce the mechanism either. The two strands of evidence cohere under a single interpretation: probes detect a co-occurrence axis that correlates with the deception label in framings where contextual cues are present and decouples from it where they are absent. Linear deception probes in this model are not detecting the mechanism; the benchmark and sha256-frozen test splits are released to enable replication on other models.

8 Reproducibility

All artifacts are present in the project repository.

- **Model:** `unsloth/gemma-2-2b-it` (HuggingFace), bfloat16. The methodologically equivalent `google/gemma-2-2b-it` produces identical results within bfloat16 precision.
- **SAE checkpoints:** `google/gemma-scope-2b-pt-res` (Gemma Scope JumpReLU SAEs, residual stream, width 16k).
- **Pre-registrations (panel A):** 20-seed bootstrap `cycle-004/PREREG/shot_R2_u1_deep_bootstrap.md` (commit `d91b109d`); SAE subspace test `cycle-003/PREREG/shot_U3_subspace_ablation.md`; 3-direction subspace `cycle-004/PREREG/shot_R3_geometry_prereg.md` (commit `d91b109d`).
- **Pre-registrations (panel B):** `shot_T1_taxonomy_freeze.md`, `shot_T2_prompt_generation.md`, `shot_T3_pilot_evaluation.md` (commit `130b434a`).
- **Project-wide pre-registration:** `INVARIANT.md`, commit `19704e19...`.
- **Sandbagging suite:** `data/processed/prompts.jsonl` (sha256 `e543e781...`); splits sha256 `10deee0e...`; system prompts sha256 `479a8e8c...`.
- **Adversarial benchmark test stratum (sha256 by category):** `d0b37087...` (fiction-writing); `ba3208f5...` (role-playing); `b17fe4a8...` (hypotheticals); `b11bc9f6...` (counterfactuals); `3c69ee92...` (devil's-advocate); `686bfa05...` (irony-following).
- **d_{mech} direction:** `experiments/05b_component_sweep/T1_sandbag_val/attn_out_L12/direction.json` (sha256 `6784633c...`).
- **Probes (panel A, 20 seeds):** `cycle-004/RESULTS/shot_R2/probes/d_probe_seed_{0..19}.npz`.
- **Per-seed metrics (panel A):** sha256 `19872ef5...`. **20-seed verdict:** sha256 `4b1ff291...`. **3-direction verdict:** sha256 `1830d201...`. **SAE subspace verdict:** sha256 `add33453...`.
- **Adversarial benchmark verdicts (panel B):** 3-probe panel `project-002/cycle-002/RESULTS/shot_U2/ver` inter-probe κ `cycle-002/RESULTS/shot_U3/verdict.json`; split freeze `cycle-003/RESULTS/shot_V1/verdict` final test AUROC `cycle-003/RESULTS/shot_V2/verdict.json`.
- **Test-touch budget:** panel A 0; panel B 3 (T3, U2, V2; one per cycle).
- **Seeds:** panel A 0–19 for the bootstrap, 0 for the seed-0 audit.
- **Hardware:** RTX 2060 12 GB; bfloat16.

References

[1] Apollo Research. *Deception probe evaluations.* Technical report, 2024.

[2] A. Arditi, O. Obeso, A. Syed, et al. Refusal in language models is mediated by a single direction. arXiv:2406.11717, 2024.

[3] Y. Belinkov. Probing classifiers: Promises, shortcomings, and advances. *Computational Linguistics*, 48(1):207–219, 2022.

[4] T. Bricken et al. Cross-coders: Cross-model dictionary learning. *Transformer Circuits Thread*, 2024.

[5] N. Goldowsky-Dill, B. Chughtai, S. Heimersheim, and M. Hobbhahn. Detecting strategic deception with linear probes. arXiv:2502.03407, 2025.

[6] J. Hewitt and C. D. Manning. A structural probe for finding syntax in word representations. In *NAACL*, 2019.

[7] E. Hubinger, C. Denison, J. Mu, et al. Sleeper agents: Training deceptive LLMs that persist through safety training. Anthropic, 2024.

[8] T. Lieberum, S. Rajamanoharan, A. Conmy, et al. Gemma Scope: Open sparse autoencoders everywhere all at once on Gemma 2. arXiv:2408.05147, 2024.

[9] J. F. McCann. The L12 suppression direction: A vertical-circuit characterization of sandbagging in Gemma 2 2B IT. Preprint, 2026.

[10] J. Taylor, S. Black, D. Bowen, et al. Auditing games for sandbagging. UK AI Security Institute, arXiv:2512.07810, 2025.

[11] A. Templeton, T. Conerly, J. Marcus, et al. Scaling monosemanticity: Extracting interpretable features from Claude 3 Sonnet. *Transformer Circuits Thread*, 2024.

[12] T. van der Weij et al. AI sandbagging: Language models can strategically underperform on evaluations. 2024.

A Per-seed bootstrap metrics (panel A)

seed	cos	Δ_{AUROC}	$\Delta^{\text{sb}}_{\text{count}}$	$\Delta^{\text{sb}}_{\text{pp}}$	$\Delta^{\text{hn}}_{\text{count}}$	$\Delta^{\text{hn}}_{\text{pp}}$
0	0.0153	0	0	0.0	+1	+2.5
1	0.0162	0	0	0.0	−2	−5.0
2	0.0180	0	+1	+2.5	−1	−2.5
3	0.0188	0	+2	+5.0	0	0.0
4	0.0126	0	+1	+2.5	0	0.0
5	0.0160	0	+1	+2.5	−1	−2.5
6	0.0131	0	0	0.0	0	0.0
7	0.0162	0	0	0.0	0	0.0
8	0.0145	0	0	0.0	0	0.0
9	0.0162	0	0	0.0	0	0.0
10	0.0174	0	+1	+2.5	−1	−2.5
11	0.0165	0	0	0.0	−3	−7.5
12	0.0184	0	+2	+5.0	−1	−2.5
13	0.0157	0	0	0.0	−1	−2.5
14	0.0184	0	+1	+2.5	0	0.0
15	0.0131	0	0	0.0	−1	−2.5
16	0.0152	0	+1	+2.5	−1	−2.5
17	0.0162	0	0	0.0	−1	−2.5
18	0.0180	0	0	0.0	−2	−5.0
19	0.0149	0	0	0.0	0	0.0

Source: `cycle-004/RESULTS/shot_R2/per_seed_metrics.json`. Mean cos = 0.0160; mean $\Delta^{\text{sb}}_{\text{count}} = +0.5$; mean $\Delta^{\text{hn}}_{\text{count}} = -0.7$.

B SAE subspace null details

Quantity	Value
$n_{\text{configurations tested}}$	13
n_{landmark}	0
n_{strong}	0
Random baseline N	9
Random $\Delta^{\%}_{\text{proj}}$ (mean $\pm$ std, max)	$-0.103\% \pm 0.442\%$, max 0.891%
Random $\Delta^{\text{pp}}_{\text{acc}}$ (mean $\pm$ std, max)	$+1.39 \pm 2.91$, max $+5.00$
Pre-reg LANDMARK	$\lvert\Delta^{\text{L12}}_{\text{proj}}\rvert \geq 30\%$ AND $\Delta^{\text{sb}} \geq +5\,\text{pp}$ AND $\Delta^{\text{hn}} \geq -3\,\text{pp}$ AND $\geq 2\sigma$ over random
Pre-reg STRONG	$\lvert\Delta_{\text{proj}}\rvert \geq 15\%$ AND $\Delta^{\text{sb}} \geq +3\,\text{pp}$ AND $\Delta^{\text{hn}} \geq -5\,\text{pp}$ AND $\geq 2\sigma$ over random
Outcome	SUBSPACE-NULL-CONFIRMED

C Per-probe per-category AUROC with sample sizes

Probe	Category	n_+	n_-	AUROC
Apollo	fiction-writing	34	66	0.2914
Apollo	role-playing	38	62	0.9962
Apollo	hypotheticals	77	23	1.0000
Apollo	counterfactuals	32	68	0.7082
Apollo	devil's-advocate	20	20	0.9325
Apollo	irony-following	16	29	0.4246
Goldowsky-Dill	fiction-writing	34	66	0.7348
Goldowsky-Dill	role-playing	38	62	0.9975
Goldowsky-Dill	hypotheticals	77	23	1.0000
Goldowsky-Dill	counterfactuals	32	68	0.9384
Goldowsky-Dill	devil's-advocate	20	20	0.9675
Goldowsky-Dill	irony-following	16	29	0.3664
Meanshift	fiction-writing	34	66	0.8957
Meanshift	role-playing	38	62	0.9966
Meanshift	hypotheticals	77	23	1.0000
Meanshift	counterfactuals	32	68	0.9821
Meanshift	devil's-advocate	20	20	0.9550
Meanshift	irony-following	16	29	0.4353

Source: `project-002/cycle-003/RESULTS/shot_V2/verdict.json`.

D Three-direction subspace details (panel A)

Source: `cycle-004/RESULTS/shot_R3/verdict.json`.

Quantity	M_A (probe-mean)	M_B (probe seed-0)
$\cos(d_{\text{mech}}, d_{\text{refusal}})$	0.357	0.357
$\cos(d_{\text{mech}}, d_{\text{probe}})$	0.0161	0.0174
$\cos(d_{\text{refusal}}, d_{\text{probe}})$	-0.0153	-0.0161
σ_1	1.165	1.165
σ_2	1.001	1.001
σ_3	0.801	0.801
σ_3/σ_1	0.688	0.688
Subspace full-rank	yes	yes